

1) Bellow is a list of calories and cholesterol amounts in 4 randomly selected menu items from 4 different fast food companies.

<i>Company</i>	<i>Menu Item</i>	<i>Calories</i>	<i>Cholesterol (mg)</i>
McDonald's	Bacon, Egg, & Cheese Biscuit	460	215
McDonald's	Big Mac®	590	85
McDonald's	Filet-O-Fish®	390	30
McDonald's	Cheeseburger	300	40
Burger King	Whopper® Sandwich with Cheese	740	115
Burger King	Cheeseburger	280	45
Burger King	Crispy Chicken Sandwich	670	60
Burger King	Bacon, Egg & Cheese Biscuit	400	170
Wendy's	Baconator®	960	155
Wendy's	Bacon Double Stack®	440	65
Wendy's	Classic Chicken Sandwich	490	75
Wendy's	Sausage, Egg, and Cheese Biscuit	580	285
Chick-fil-A	Chicken Biscuit	460	45
Chick-fil-A	Bacon, Egg, and Cheese Biscuit	420	180
Chick-fil-A	Grilled Chicken Sandwich	390	75
Chick-fil-A	Chicken Nuggets (8 count)	250	80

a) Using your calculator, find the following summary statistics for CALORIES for each of the companies.

	Mean	Min	Q1	Med	Q3	Max	Std. Dev
McDonald's							
Burger King							
Wendy's							
Chick-fil-A							

Which company has the greatest calorie variability in the distribution? _____

b) There are many methods for determining outliers. Two methods frequently used in AP Statistics are:

Rule #1: An outlier is a value greater than $1.5 \times \text{IQR}$ above the third quartile or more than $1.5 \times \text{IQR}$ below the first quartile.

Rule #2: An outlier is a value located 2 or more standard deviations above, or below, the mean.

Using rule #1, are there any outliers in the Wendy's distribution? Show your work.

Using rule #2, are there any outliers in the Wendy's distribution? Show your work.

c) Draw four modified boxplots comparing the calories for each of the four companies.

d) In your town, McDonald's and Burger King are on the Northside and Wendy's and Chick-fil-A are on the Southside. Using your calculator, find the following summary statistics for CHOLESTEROL for the Northside and Southside fast food restaurants.

	Mean	Min	Q1	Med	Q3	Max	Std. Dev
Northside							
Southside							

Which region has the greatest cholesterol variability in the distribution? _____

e) Using rule #1, are there any outliers in the Southside’s distribution? Show your work.

Using rule #2, are there any outliers in the Southside’s distribution? Show your work.

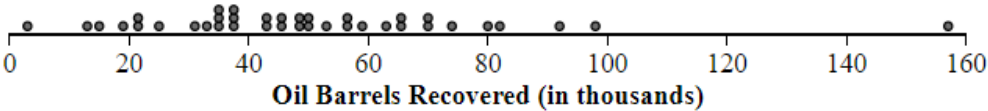
f) Remove the value of 285mg from the Southside cholesterol’s data set. Use your calculator to find the following values again:

	Mean	Min	Q1	Med	Q3	Max	Std. Dev
Southside							

What values changed the most? What values changed the least?

2) When drilling for oil wells, researchers have to take into account how much oil each well will eventually produce before deciding if they should drill in a specific location. In the table below, there are the total amounts of oil recovered from 38 wells in the Michigan basin, in thousands of barrels.

3	31	38	50	65	92
13	33	43	50	66	98
15	35	43	53	70	157
19	35	45	56	70	
21	35	46	57	74	
22	37	48	59	80	
25	37	49	63	82	



a) What measures would you use to describe the center and spread of these data? Justify your answer.

b) Find the five number summary for these data.

c) Are there any outliers? Justify your answer.

d) Draw a modified boxplot of this distribution.

e) When looking at the boxplot of the distribution, how can you tell (without doing any calculations) that the mean of these data is larger than the median?

3) A student studying the sleeping habits of seniors at his school asked 36 randomly-selected seniors how many hours of sleep they got the previous night. The data, rounded to the nearest half-hour, is given in the table below.

8	7.5	9	7.5	9	6	5	9	7.5	7	8	7
6.5	8.5	8	6.5	8.5	6	7	7.5	7	6	8.5	7
7	8	7	7.5	7	6	7	8	7.5	6	7	5

a) Using your calculator, find the mean and median of this data. Without graphing it, what shape do you suspect, give the values of the mean and median?

b) Find and interpret the standard deviation.

c) Suppose 4 more values were added to the data, each exactly equal to the mean. Would this have any impact on the standard deviation? Explain, without using any calculations.

_____ 4) In a distribution of 3000 scores, the mean is 78 and the median is 95. One would expect this distribution to be:

- (A) skewed to the right
- (B) skewed to the left
- (C) symmetrical and mound-shaped
- (D) symmetrical and uniform
- (E) bimodal

_____ 5) The five-number summary for a one-variable dataset is {5, 18, 20, 40, 75}. If you wanted to construct a modified boxplot for the dataset, what would be the maximum possible length of the right side “whisker”?

- (A) 35
- (B) 33
- (C) 5
- (D) 55
- (E) 73

_____ 6) Which of the following are true statements?

- I. The standard deviation is the square root of the variance.
- II. The standard deviation is zero only when all values are the same.
- III. The standard deviation is strongly affected by outliers.

- (A) I and II
- (B) I and III
- (C) II and III
- (D) I, II, and III
- (E) None of the above gives the complete set of true responses.

_____ 7) In order to rate TV shows, phone surveys are sometimes used. Such a survey might record several variables, some of which are listed below. Which of these variables are categorical?

- I. The type of show being watched
- II. The number of persons watching the show
- III. The ages of persons watching the show
- IV. The name of the show being watched
- V. The number of times the show has been watched in the last month

- (A) II, III, and V
- (B) I only
- (C) I and V
- (D) I and IV
- (E) None of the above describes the complete set of correct responses